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Roll No

AU-7004(1)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Alternative Fuels and Pollution Control

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) How are SI and CI engine fuels rated?
b) Explain the reason for looking for alternative fuels.
2. a) Discuss the modification required to use the alternate fuels in SI engine.
b) Discuss the process of flue gas analysis with neat sketch.
3. a) Discuss various types of solar energy collectors with neat sketch and write the use of each.
b) Write short notes.
 - i) Wind energy
 - ii) Bio energy
4. a) Explain with neat sketch the biodiesel production process.
b) Write note on any two parts:
 - i) Ethanol
 - ii) Storage of hydrogen fuel
 - iii) Biogas

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5.
 - a) Explain in brief the working of catalytic converter and also its types.
 - b) Explain the need and working of EGR system for controlling engine temperature.
6.
 - a) Explain working of evaporative control system for PFI engine with neat sketch.
 - b) Explain with neat sketch constructional and operational features of NDIR analyzer for measurement of CO and HC concentration.
7.
 - a) Compare the properties Petrol, CNG and LPG.
 - b) Discuss in detail wind power generation.
8. Discuss any two:
 - i) Geo thermal energy
 - ii) Hydrogen energy
 - iii) Fuel cells.
